

Dataset Design & Quality Expectations

Optical Camera Communication (OCC) Video Message Decoding

1. Expected Files, Metadata, and Targets

Expected input files

- Raw OCC video files (.mp4 or similar), one or more per transmitted message, OOK-encoded.
- Possibly a manifest/index file (e.g. manifest.csv) listing filename, fps, resolution, and any known label per video.
- Ground-truth labels where available: transmitted bit sequence and/or decoded message text — could arrive as a separate labels file rather than embedded in the video.

Important metadata to check per video

- Frame rate (fps) and whether it is constant within the file.
- Resolution and rolling-shutter vs. global-shutter capture type, if stated.
- Duration and total frame count (used to cross-check reported vs. actual fps).
- Capture conditions if noted: distance, lighting, indoor/outdoor, camera motion.

Prediction target

- Primary target: the transmitted binary sequence recovered from the video.
- Secondary target: the final decoded message reconstructed from that sequence.
- If no ground truth exists for a given video, it can still be used for qualitative/robustness analysis but not for BER-style scoring.

2. Assumed Folder / Schema Convention

- raw/ — untouched original video files exactly as supplied, plus a manifest of filename/fps/resolution/known-label. This folder is never modified after initial receipt.
- processed/ — derived artifacts only: extracted frames, ROI masks, and extracted intensity-signal time series. Fully regenerable from raw/ by the preprocessing pipeline.
- labels/ — ground-truth bit sequences / decoded message text, kept separate from raw/ so labels can be corrected without touching source video.
- splits/ — train / validation / test file lists, defined by message/session identity rather than by individual file (see leakage risk below).

3. What a Correct Train / Validation / Test Split Must Preserve

- No video (or near-duplicate/repeat capture of the same message) appears in more than one split.
- Each split contains a representative mix of capture conditions (lighting, distance, motion) rather than all 'easy' videos in train and all 'hard' videos in test.
- The test split is only touched once, at final evaluation — not used repeatedly during threshold/parameter tuning.
- If very few videos are supplied overall, validation may need to rely on cross-validation or leave-one-video-out rather than a fixed three-way split — to be confirmed once actual dataset size is known.

4. Dataset-Risk Checklist

Risk Category	What It Would Look Like Here	Why It Matters	Planned Check
Missing / corrupt data	A video file that won't open, is truncated, or has 0-length frames.	Would silently break frame extraction or produce garbage intensity signal.	Verify every file opens and has non-zero, expected frame count before any processing.
Annotation quality	Ground-truth bit sequence / message text missing, mislabeled, or inconsistent with the actual video content.	Any decoder evaluated against wrong ground truth looks wrong even if it isn't.	Spot-check labels against a manually-decoded sample of 23 videos before trusting them at scale.
Class / condition imbalance	Most videos captured under similar lighting/distance, few edge cases (low light, motion, long range).	A pipeline tuned only on 'easy' videos won't generalize, and robustness claims would be unsupported.	Tabulate videos by condition (lighting, distance, motion) once metadata is available; flag underrepresented conditions.
Temporal / sequence risk	Frame drops or variable frame rate within a single video, or inconsistent fps across videos.	Breaks the fixed relationship between frame index and time, which synchronization depends on.	Check reported fps against actual frame count / duration per file; do not assume fps is constant without verifying.
Leakage	The same physical message/video (or a near duplicate/repeat) appearing in both train and test splits.	Inflates apparent decoding accuracy without proving generalization.	Split by underlying message/session identity, not by file, if multiple files share one transmission.

5. Next Step

This note captures expected structure and risks prior to final-dataset inspection. Once the dataset is confirmed usable, Day 7's raw-data inspection will validate or correct every assumption above against the actual supplied files.